Search & Rescue LoRa Node - Hardware Specification & Wiring Manual

SoC: ESP32-C6FH4 (RISC-V) Board: Seeed Studio XIAO ESP32C6 Firmware: Meshtastic Custom HAL

> **Hardware Compatibility Verified:** This hardware profile has been tailored for the **Seeed Studio XIAO ESP32C6** board. By routing SX1262 RESET to the hardware reset bus and running the NEO-6M GPS in passive receive mode, exactly 11 components are routed to the 11 edge pins with zero pin contention and zero strapping pin conflicts.

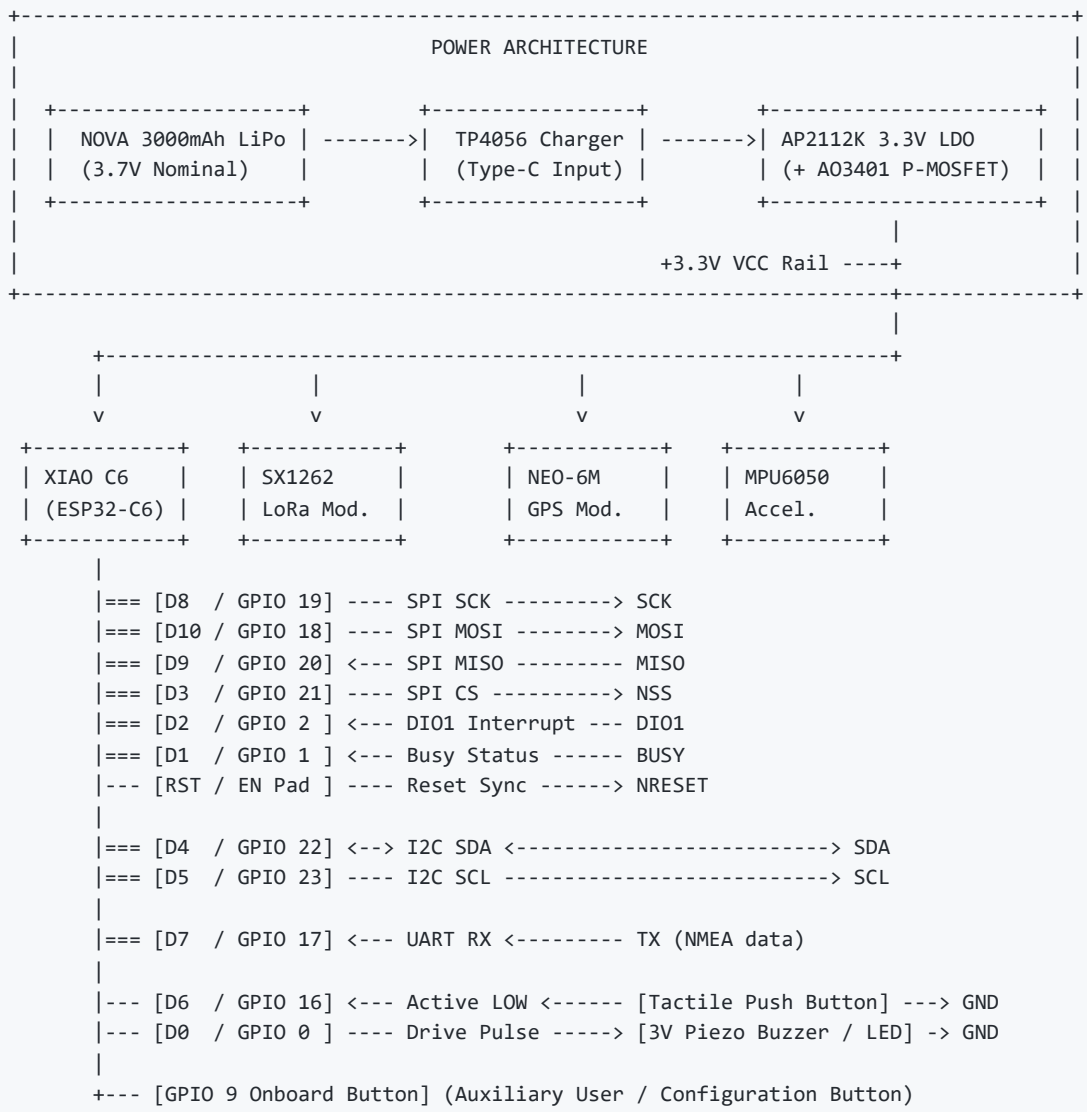
1. Bill of Materials (BOM) Breakdown

Part Description	Model / Specification	Interface	Function in System
Main Microcontroller	Seeed Studio XIAO ESP32C6 (ESP32-C6FH4)	USB-C / 11 GPIO Headers	Core processor, FreeRTOS, BLE 5, Wi-Fi 6
LoRa Transceiver	IndieSemiC ISC-SX1262-B (SX1262)	SPI (SCK, MOSI, MISO, CS, DIO1, BUSY)	Long-range mesh communication (868 MHz)
GPS Navigation Unit	u-blox NEO-6M GPS Module	UART RX (Passive 9600 baud)	Positioning & satellite coordinates
Inertial Sensor	MPU6050 / MPU6500 Breakout	I2C (SDA, SCL)	Fall detection & motion tracking
Emergency Push Button	Tactile Waterproof Push Button (6x6mm)	Digital Input (GPIO 16 / Active LOW)	Instant SOS broadcast & node wakeup
Acoustic / Visual Beacon	3V SMD Piezo Buzzer & Indicator LED	Digital Output (GPIO 0)	Audible SAR beacon & message alerts
Rechargeable Battery	NOVA 105050 3000mAh 3.7V LiPo	Battery Leads	Long-duration autonomous power source
Charging Controller	TP4056 Type-C Charger Module	USB-C & Battery Out	Safe LiPo charging & battery protection
Power Management	AO3401 P-MOSFET + AP2112K LDO	Power Rail Distribution	Ultra-low dropout 3.3V rail & load switching
LoRa Antenna System	865–868 MHz Stub Antenna (50mm) + IPEX to SMA	RF IPEX / SMA Female	Mesh transmission and reception
GPS Antenna System	Active 25x25mm Ceramic Patch Antenna	U.FL / Coaxial	High-sensitivity satellite reception

2. Complete 11-Pin Wiring Map (Seeed XIAO ESP32C6)

XIAO Pin	ESP32-C6 GPIO	Target Component	Function	Physical Wiring Detail
D0	`GPIO 0`	**Piezo Buzzer & LED**	Beacon Alert	Connect Buzzer (+) & LED Anode to D0; Cathodes to GND
D1	`GPIO 1`	**SX1262 LoRa**	BUSY Status	Connect directly to SX1262 `BUSY` pin
D2	`GPIO 2`	**SX1262 LoRa**	DIO1 Interrupt	Connect directly to SX1262 `DIO1` pin
D3	`GPIO 21`	**SX1262 LoRa**	SPI NSS / CS	Connect directly to SX1262 `NSS / CS` pin
D4	`GPIO 22`	**MPU6050**	I2C SDA	Connect directly to MPU6050 `SDA` pin
D5	`GPIO 23`	**MPU6050**	I2C SCL	Connect directly to MPU6050 `SCL` pin
D6	`GPIO 16`	**Tactile Button**	SOS / User Button	Connect one pin to D6, other pin to GND (Active LOW)
D7	`GPIO 17`	**NEO-6M GPS**	UART RX	Connect to NEO-6M `TX` pin (GPS RX left floating)
D8	`GPIO 19`	**SX1262 LoRa**	SPI SCK	Connect directly to SX1262 `SCK` clock pin
D9	`GPIO 20`	**SX1262 LoRa**	SPI MISO	Connect directly to SX1262 `MISO` pin
D10	`GPIO 18`	**SX1262 LoRa**	SPI MOSI	Connect directly to SX1262 `MOSI` pin
RST Pad	`RST / EN`	**SX1262 LoRa**	Hardware Reset	Connect SX1262 `RESET` to XIAO RST pad (Auto-reboot sync)

3. System Interconnect & Architecture Diagram



4. Key Hardware Engineering Bypasses

1. LoRa Reset Line Optimization (Hardware Sync):
Standard LoRa modules require a reset pulse on startup. By tying the SX1262 `NRESET` pin directly to the XIAO's hardware `RST / EN` pad, the LoRa transceiver is guaranteed a clean hardware reset simultaneously whenever the ESP32-C6 resets or wakes up. In the firmware, this is configured as `#define LORA_RESET RADIOLIB_NC`, freeing an entire GPIO.

2. GPS Unidirectional Telemetry Mode:
The u-blox NEO-6M continuously streams standard NMEA positioning sentences (GPGGA, GPRMC) at 9600 baud as soon as power is applied. The node only needs to receive these sentences. By leaving the NEO-6M RX pin floating and connecting only its TX pin to XIAO D7 (`GPIO 17`), a second full GPIO pin is conserved without sacrificing GPS functionality.

> **3. Boot Strap Safety:**
> ESP32-C6 strapping pins (GPIO 8 and GPIO 9) control ROM booting and flashing. On the Seeed Studio XIAO ESP32C6, the edge pins (D0–D10) route to non-strapping pins. The onboard physical button is safely pulled up to 3.3V on GPIO 9 and remains available as a secondary user button.

5. Firmware Target & Direct Flashing Instructions

The custom firmware variant is integrated directly into the Meshtastic build system:

- **Variant Path:** `firmware/variants/esp32c6/diy/rescue_node_c6/`
- **Build Environment:** `rescue_node_c6`
- **Detected Serial Port:** `COM7`

Command-Line Build & Flash (via PowerShell):

```
# 1. Compile the custom rescue_node_c6 firmware:
uvx platformio run -e rescue_node_c6

# 2. Upload directly to the connected ESP32-C6 on COM7:
uvx platformio run -e rescue_node_c6 --target upload --upload-port COM7

# 3. Open real-time serial monitor (460800 baud):
uvx platformio device monitor -b 460800 -p COM7
```
